SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: *Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)*

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Industry Problem Solving Task

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Annexure A

Industry Problem Solving Task

1: Automated Defect Detection in Manufacturing

A precision engineering company wants to automatically detect defects in mechanical components using computer vision. Parts vary slightly in shape due to manufacturing tolerances.

Questions:

1. How can PCA-based shape priors and contour-based representations be used to model standard component shapes?
2. Propose a pipeline using machine learning (SVM, GMM) or deep learning (CNN) to classify defective vs. non-defective parts.
3. Discuss methods to handle noise, partial occlusions, and varying illumination in industrial images.

2: Autonomous Vehicle Obstacle Recognition

A self-driving car company needs to detect rare or unexpected obstacles on urban roads. The labeled data for these obstacles is limited.

Questions:

1. Explain how data augmentation using GANs can help generate synthetic images for training.
2. Compare the effectiveness of CNNs versus Vision Transformers for detecting small or sparse objects in high-resolution images.
3. Suggest a real-time inference pipeline for obstacle detection, considering computational constraints in embedded systems.

1.Automated Defect Detection in Manufacturing

1. Understanding of the Industry Problem

A precision engineering company needs an automated computer vision system to identify defective and non-defective mechanical components. Since manufactured parts may vary slightly because of manufacturing tolerances, the system should not classify every small shape variation as a defect.

The major challenges are:

- Small variations in component shape
- Surface noise and scratches
- Partial occlusion
- Different illumination conditions
- Different orientations and positions of parts
- Need for accurate and reliable defect detection

The proposed system should learn the **normal shape and appearance of components** and identify deviations that indicate defects.

2. PCA-Based Shape Priors and Contour-Based Representation

PCA-Based Shape Priors

Principal Component Analysis (PCA) can be used to learn the normal variation in component shapes from a set of defect-free training images.

The process is:

Training Images → Extract Contours → Align Shapes → PCA → Mean Shape + Shape Variations

Each component contour is represented using boundary points. PCA learns the main variations in these points.

The shape can be represented as:

$x=\bar{x}+Pb$

Where:

- x = reconstructed component shape
- $\bar{x}$ = mean shape
- P = principal components
- b = shape parameters

The PCA model allows small manufacturing variations while identifying large deviations as possible defects.

Contour-Based Representation

The boundary of each mechanical component can be extracted using edge detection techniques such as Canny edge detection.

Important contour features include:

- Boundary coordinates
- Area
- Perimeter
- Curvature
- Shape moments
- Circularity

The contour of a new component is compared with the learned standard shape. Large differences can indicate a defective part.

3. Proposed Defect Detection Pipeline

Pipeline

Image Acquisition → Preprocessing → Segmentation → Contour Extraction → Feature Extraction → PCA Shape Model → SVM/CNN Classification → Defect Detection → Evaluation

Step 1: Image Acquisition

Capture images of mechanical components using industrial cameras under controlled conditions.

Step 2: Preprocessing

Noise and illumination variations are reduced using:

- Gaussian/median filtering
- Image normalization
- Contrast enhancement
- Background removal

Step 3: Segmentation

Separate the mechanical component from the background using thresholding or other segmentation techniques.

Step 4: Feature Extraction

Extract:

- Shape features
- Contour features
- Texture features
- Edge information

Step 5: Classification

Two possible approaches can be used.

SVM:
Extracted features are given to an SVM classifier to classify the component as:

- **Defective**
- **Non-defective**

GMM:
A Gaussian Mixture Model can learn the probability distribution of normal component features. Features with very low probability can be treated as potential defects.

CNN:
A CNN can directly learn image features and classify components using labeled training images.

Step 6: Final Decision

The system produces:

Component → Defective / Non-defective

A bounding box or highlighted region can also be displayed to indicate the defect location.

4. Handling Noise, Occlusion and Illumination

Noise

Noise can be reduced using:

- Median filtering
- Gaussian filtering
- Morphological operations

Partial Occlusion

Partial occlusions can be handled using:

- PCA shape priors
- Partial contour matching
- Data augmentation
- Robust feature extraction

The learned shape prior can estimate the expected complete component shape even when a portion is hidden.

Varying Illumination

Illumination problems can be reduced using:

- Histogram equalization
- Adaptive histogram equalization
- Image normalization
- Controlled industrial lighting
- Brightness and contrast augmentation

5. Analysis and Performance Evaluation

The system should be evaluated using a separate test dataset containing both defective and non-defective components.

Important performance measures are:

Accuracy=TP+TN+FP+FNTP+TN Precision=TP+FPTP Recall=TP+FNTP F1=Precision+Recall2(Precision)(Recall)

A confusion matrix can also be used to analyze correct and incorrect classifications.

For industrial applications, high recall is particularly important because missing a defective component can result in safety and quality problems.

6. Professional and Practical Solution

A practical industrial system can combine PCA-based shape modeling with CNN/SVM classification. PCA can handle normal manufacturing shape variations, while the classifier can learn visual differences between defective and non-defective parts.

Final Conclusion

PCA-based shape priors and contour representations provide a reliable way to model the normal shape of mechanical components. Machine learning or deep learning models can then classify defects based on shape, texture, and appearance. Proper preprocessing, illumination control, data augmentation, and performance evaluation make the proposed system suitable for automated industrial inspection.

2.Autonomous Vehicle Obstacle Recognition

1. Understanding of the Industry Problem

A self-driving car must detect rare and unexpected obstacles on urban roads. Examples include traffic cones, animals, unusual objects, and partially visible obstacles.

The main challenge is that only a small number of labeled images may be available for rare objects. Therefore, the system needs to:

- Generate additional training data
- Detect small and distant objects
- Work in complex urban environments
- Provide accurate predictions
- Operate in real time on embedded hardware

2. GAN-Based Data Augmentation

A Generative Adversarial Network (GAN) can generate realistic synthetic images of rare obstacles.

A GAN consists of two main networks:

Generator

The generator creates synthetic obstacle images from random input.

Discriminator

The discriminator tries to distinguish between real and generated images.

The two networks are trained together until the generated images become realistic.

Process

Real Obstacle Images → GAN Training → Synthetic Images → Data Augmentation → Object Detection Model

For example, if only a small number of traffic-cone images are available, GANs can generate additional images with variations in:

- Position
- Scale
- Rotation
- Background
- Lighting
- Viewing angle

These synthetic images are added to the real training dataset.

Benefit

GAN-based augmentation helps reduce the effect of class imbalance and improves the ability of the detection model to recognize rare obstacles.

3. CNN vs Vision Transformer

Feature	CNN	Vision Transformer (ViT)
Feature extraction	Uses convolution operations	Uses self-attention
Local features	Very effective	Effective
Global relationships	More limited	Very effective
Small objects	Can perform well with multi-scale features	Can capture relationships across image regions
Training data	Generally works well with smaller datasets using transfer learning	Often benefits from large datasets or pretraining
Computational requirement	Usually lower	Usually higher
Embedded deployment	More suitable for resource-limited devices	More challenging but possible with optimization
High-resolution images	Efficient with suitable detection architectures	Strong global context but computationally expensive

Analysis

CNNs are highly suitable when computational resources are limited because they can efficiently extract local features such as edges, shapes, and textures.

Vision Transformers use self-attention to understand relationships between different regions of an image. This can be useful when a small obstacle needs to be distinguished from a complex urban background.

However, because the available labeled dataset is small, a **pretrained CNN or pretrained Vision Transformer** should be used rather than training a large model from scratch.

For a real-time embedded self-driving system, a lightweight CNN-based detector is generally a practical starting point. A Transformer-based detector can be considered when sufficient computational resources and suitable optimization are available.

4. Real-Time Inference Pipeline

A practical real-time obstacle detection system can use the following pipeline:

Camera → Image Preprocessing → Lightweight Detection Model → Non-Maximum Suppression → Obstacle Classification → Bounding Box → Decision System

Step 1: Camera Input

The vehicle camera continuously captures RGB images or video frames.

Step 2: Preprocessing

Frames are:

- Resized
- Normalized
- Converted into the required input format

Step 3: Lightweight Detection Model

A lightweight CNN-based object detector is used to detect obstacles quickly.

The model produces:

- Object class
- Bounding box
- Confidence score

Step 4: Post-Processing

Non-Maximum Suppression (NMS) removes duplicate bounding boxes.

Step 5: Obstacle Decision

If a rare obstacle is detected with sufficient confidence, the system sends the information to the vehicle's decision-making module.

For example:

Animal detected → Obstacle warning → Slow down / Avoid obstacle

Step 6: Embedded Optimization

To achieve real-time performance, the model can be optimized using:

- Model quantization
- Pruning
- Smaller input resolution where appropriate
- Hardware acceleration
- Efficient network architectures
- Batch size of 1 for streaming inference

5. Analysis and Performance Evaluation

The obstacle detection system should be evaluated using:

Precision

Measures how many detected obstacles are actually correct.

Precision=TP+FPTP

Recall

Measures how many actual obstacles are successfully detected.

Recall=TP+FNTP

F1-Score

F1=Precision+Recall2(Precision)(Recall)

IoU

Intersection over Union measures the overlap between the predicted and actual bounding boxes.

IoU=Area of UnionArea of Intersection

mAP

Mean Average Precision (mAP) can be used to evaluate overall object detection performance.

For autonomous vehicles, high recall is very important because failing to detect a rare obstacle can create a serious safety risk.